```
                              set_pg_strategy

NAME
       set_pg_strategy
              Specifies the power ground network creation strategy.

SYNTAX
       status set_pg_strategy
              strategy_name
              [-core | -design_boundary | -voltage_areas voltage_areas |
               -polygon {polygon_area} | -macros macro_names |
               -blocks block_names | -pg_regions pg_region_names]
              -pattern pattern_expr
              [-extension {extension_spec}]
              [-blockage {blockage_spec}]
              [-tag tag_name]

   Data Types
       strategy_name           string
       voltage_areas           collection
       polygon_area            list
       macro_names             collection
       block_names             collection
       pg_region_names         list
       pattern_expr            specification
       extension_spec          specification
       blockage_spec           specification
       tag_name                string

ARGUMENTS
       The options -core, -design_boundary, -voltage_areas, -polygon, -macros,
       -blocks and -pg_regions are mutually exclusive. One  and  only  one  of
       these options needs to be specified.

       strategy_name
              Specifies the name of the strategy.

       -core  Uses the core area as the power ground region.

       -design_boundary
              Uses the design boundary as the power ground region.

       -voltage_areas voltage_area
              Uses the specified voltage area as the power ground region.

       -polygon {polygon_area}
              Uses  the specified polygon area as the power ground region. The
              polygon area is specified by a list of coordinate points.

       -macros {macro_names}
              Uses the specified macros as the power ground region.

       -blocks {block_names}
              Uses the specified blocks as the power ground region.

       -pg_regions pg_region_names
              Uses the power ground regions created  by  the  create_pg_region
              command  as  the  power ground regions.  Multiple regions can be
              specified in the list pg_region_names.

       -pattern pattern_expr
              Specifies the power ground pattern to be  instantiated  in  this
              strategy.  The pattern can be wire, composite, mesh, ring, macro
              connection, standard cell rail connection, or  special  pattern.
              Only one pattern can be specified and this is a required option.

              For  wire,  composite,  mesh, connection or special pattern, the
              strategy instantiates the pattern inside  the  routing  area  by
              repeating the pattern.  For macro connection pattern, the strat-
              egy instantiates the pattern by applying the pattern  to  macros
              inside  the  routing  area.  For a standard cell rail connection
              pattern, the strategy instantiates the pattern by creating stan-
              dard cell rails inside the routing area.  The pattern expression
              pattern_expr is specified as follows:

              {name: pattern_name}
              {nets: real_net_list}
```

{parameters: var_list}
{offset: {x_offset y_offset}}
{offset_start: boundary | {x y}}

The name keyword followed by pattern_name specifies the name of the pattern. For standard cell connection pattern, if the specified pattern name is std_cell_rail and this pattern does not exist, a default standard cell rail pattern std_cell_rail is created.

The nets keyword followed by real_net_list specifies the list of net names to be mapped into the symbolic list of nets in pattern_name.

The parameters keyword followed by var_list specifies the parameter list. The parameter list is passed as arguments into the pattern pattern_name.

The offset keyword followed by {x_offset y_offset} specifies the horizontal and vertical offsets when creating the first pattern. By default, the offset is 0 for both horizontal and vertical offsets. Offset does not apply to the scattered pin macro connection pattern or the standard cell rail pattern.

The offset_start keyword specifies the offset reference point in routing area for pattern offsets. The boundary keyword refers to the lower-left point of the routing area bounding box. Any other point can also be specified as the reference point by coordinate {x y}. By default, boundary is used. Offset start does not apply to scattered pin macro connection pattern or standard cell rail pattern.

For a ring pattern, the strategy instantiates the pattern around the routing area. The pattern expression pattern_expr is specified as follows:

```
{name: pattern_name}
{nets: real_net_list}
{parameters: var_list}
{offset: {x_offset y_offset}}
{skip_sides: id_list}
{side_offset: side_offset_spec}
```

The pattern keyword followed by pattern_name specifies the pattern name.

The nets keyword followed by real_net_list specifies the net name list. The nets are mapped into the symbolic list of nets in pattern_name.

The parameters keyword followed by var_list specifies the parameter list which will be passed as arguments into the pattern pattern_name.

The offset keyword followed by {x_offset y_offset} specifies the horizontal and vertical offsets when creating ring segments around the routing area. The offset is the distance between innermost ring to the routing area boundary. By default, the offset is 0 for both horizontal and vertical offsets.

The skip_sides keyword specifies the edge ID list to be skipped. id_list specifies the edge ID list to be skipped for ring creation. Each edge is indexed by a unique number. Edge numbering starts from 1 at the leftmost vertical edge. If there is more than one leftmost edge, the bottommost edge is the starting edge. The edge number increases by 1 as you proceed around the shape in the clockwise direction. By default, no edge is skipped for ring creation.

The side_offset keyword specifies the offset for each edge. Each edge is indexed by a unique number. Edge numbering starts from 1 at the leftmost vertical edge. If there is more than one leftmost edge, the bottommost edge is the starting edge. The edge number increases by 1 as you proceed around the shape in the clockwise direction. side_offset_spec specifies the offset for each edge as a pair of edge_id and offset keywords. An example is as follows:

```
{{side: edge_id}{offset: edge_offset}}
```

where side and offset specify the edge ID and edge offset respectively. The offset unit is defined in the technology file. You can specify negative offsets to shrink the region. If you do not specify offsets for all edges, the offset for the remaining edges is either 0 or the uniform offset for all edges set by -offset option.

-extension {extension_spec}

Specifies how a group of power and ground nets should extend beyond the power grid. You can specify multiple extension specifications separated by braces. Each specification contains the keywords nets, side, layers, direction, stop and stop_layers as follows:

```
{{nets: nets}
 {side : id_list}
 {layers : layer_list}
 {direction: directions}
 {stop: stop_target}}
 {stop_layers : layer_list}
```

The net names following the nets keyword are a subset of the nets specified by -nets in pattern_expr from -pattern option. If you do not specify the nets keyword, the extension applies to all the nets in the strategy.

The side keyword specifies the ring segment IDs or macro side IDs by id_list. Each ring edge or macro boundary is indexed by a unique number. Edge numbering starts from 1 at the leftmost vertical edge. If there is more than one leftmost edge, the bottommost edge is the starting edge. The edge number increases by 1 as you proceed around the shape in the clockwise direction. If not specified, the extension applies to all ring edges or all macro boundary sides. This keyword can be only applied to ring or macro connection type pattern.

The direction keyword specifies the extension direction. directions contains one or more of the letters "L", "R", "T" or "B"; "L" refers to left, "R" refers to right, "T" refers to top, and "B" refers to bottom. If this entry is not specified, the extension applies to all directions.

The layers keyword specifies the extension specifications only applied to the particular layer list by layer_list, which contains a list of routing layer names. If this keyword is not specified, the extension applies to all layers.

The specification following the stop keyword can be a distance specified in microns, a collection or a list of PG pins, or one of the following keywords: core_boundary, first_target, innermost_ring, outermost_ring, pad_ring, half_space, design_boundary, and design_boundary_and_generate_pin. core_boundary specifies the core area boundary. first_target specifies the first wire segment of the same net. collection_of_pins specifies the collection of pins to be extended to. innermost_ring specifies the innermost ring of the same net, outermost_ring specifies the outermost ring of the same net, pad_ring specifies the pad ring segment or pad ring pins of the same net, half_space specifies half spacing rule to the specified region of the strategy. and design_boundary specifies the boundary of the current design. design_boundary_and_generate_pin specifies extension to the design boundary and generating pins. If you specify a distance, the shape extends by the distance from the boundary of the routing region if the pattern is wire pattern, composite pattern, mesh pattern, or standard cell connection pattern, or from the corner of ring segments for ring pattern, or from macro pins for macro connection pattern. You can also make the shapes to extend to a pre-specified set of macro pins. There are two ways to specify the set of pins, use a list of pin names or a collection of pins. For this stop option to work, both nets and layers options need to be specified as well.

Collection of pins example set gnd_pins [get_pins -all -of_objects [get_cell $powerSwitches] -filter "name==GND"] set_pg_strategy -extension {{{nets: GND} {stop: $gnd_pins} ...}...}...

List of pin names example set gnd_pinNames "GND0 GND1 GND2 ..." set_pg_strategy -extension {{{nets: GND} {stop: $gnd_pinNames} ...}...}...

The stop_layers keyword is used together with stop: first_target, specifying that the extension specification is only applied to the particular layer list by layer_list, which contains a list of routing layer names. If this keyword is not specified, the stop: first_target extension applies to all layers.

-blockage {blockage_spec}
: Specifies the routing blockage in the power ground network. You can specify multiple blockages within the braces, one blockage per group. Blockages are separated by braces and contain the keywords nets:, layers:, and a target specification as follows:

```
{{nets: nets} {layers: layers} {target}
```

where target is a voltage area, block, macro, macro with hard keep-out-margin (KOM), PG region, hard placement blockage, or polygon that is specified as follows:

```
{voltage_areas : voltage_areas | macros : macro_names |
macros_with_keepout : macro_names | placement_blockages : all |
polygon : {polygon_area} | blocks : blocks |
pg_regions : region_list }
```

The net names following the nets keyword are a subset of the nets specified by -nets in pattern_expr from -pattern option. If you do not specify the nets keyword, the blockage applies to all the nets in the strategy. The layer names following the layers keyword are the routing layers on which power or ground straps are blocked. If you do not specify the layers keyword, the blockage applies to all routing layers. Use one of the voltage_areas, macros, polygon, blocks or pg_regions keywords to represent the blockage area. polygon_area is specified as a list of coordinate points.

-tag tag_name
: Specifies a tag name to assign to all new vias and shapes created within this strategy. This tag name is used as contents of an user attribute tag for filtering collections at later stages. By default, no tags are assigned.

DESCRIPTION

This command specifies the strategy for creating power ground network elements. The command specifies the power ground pattern, power ground nets, power ground routing area, the net extension strategy and the blockage area.

For wire, composite, mesh and special patterns, the strategy instantiates the pattern inside the routing area by repeating the pattern. For a ring pattern, the strategy instantiates the pattern around the routing area. For macro connection pattern, the strategy instantiates the pattern by applying the pattern to macros inside the routing area. For standard cell rail connection pattern, the strategy instantiates the pattern by creating standard cell rails inside the routing area. Blockage and extension can be further specified.

EXAMPLES

The following example defines a power ground strategy s1 for a composite pattern comp_pattern. Parameters 5 and 3 are passed into the pattern for wire width and spacing respectively. The routing area is inside the voltage area VA1. The real net names are VDD and VSS. Both nets extend to the outermost VDD and VSS rings. Macro hm1 inside this voltage area is treated as blockage for VDD but not for VSS.

```
  prompt> set_pg_strategy \
    s1 \
    -voltage_areas {VA1} \
    -pattern {{name: comp_pattern}{nets:{VDD VSS}}{parameters:{5 3}}}\
    -blockage {{nets: VDD}{macros: rf0/x0/t_al}}\
    -extension {stop: outermost_ring}
```

The following example defines a power ground strategy s2 for a ring pattern ring_pattern. Parameters 4, 1, 5, 2 are passed into the pattern for horizontal width, spacing, and vertical width, spacing respectively. The ring is around the routing area defined by power ground region r1. The net names are Vdd and Vss. Horizontal and vertical offsets are 10 and 20 respectively. The ring segment with edge ID 1 is extended in the top direction to the pad ring.

```
  prompt> set_pg_strategy \
    s2 \
    -pg_regions {r1} \
    -pattern {{name: ring_pattern}{nets:{Vdd Vss}} \
              {parameters:{4 1 5 2}}{offset:{10 20}}} \
    -extension {{side: 1}{direction: T}{stop: pad_ring}}
```

The following example defines a power ground strategy s3 for a ring pattern ring_pattern. The ring is around the routing area defined by power ground region r1. The net names are Vdd and Vss. Horizontal and vertical offsets are 10 and 20 respectively. The ring segment with edge IDs 1 and 2 is extended in the both bottom and right directions to the design boundary with pin created, and edge ID 3 in the top direction to design boundary with pin created.

```
prompt> set_pg_strategy \
  s3 \
  -pg_regions {r1} \
  -pattern {{name: ring_pattern}{nets:{Vdd Vss}} \
            {offset:{10 20}}} \
  -extension {{{side: 1 2}{direction: B R }{stop: design_boundary_and_generate_pin}} \
              {{side: 3}{direction: T}{stop: design_boundary_and_generate_pin}}}
```

The following example defines a power ground strategy s4 for a macro connection pattern macro_pattern. The nets are Vdd and Vss. The macros to be connected are hm1 and hm2. For the pins at boundary sides indexed by 1, 2 and 3, the extension stops at the first target. For the pins at the side indexed by 4, the extension stops at the innermost ring.

```
prompt> set_pg_strategy \
  s4 \
  -macros "$macro1 $macro2" \
  -pattern {{name: macro_pattern}{nets:{Vdd Vss}}} \
  -extension {{{side: 1 2 3}{stop: first_target}} \
                {{side: 4}{stop: innermost_ring}}}
```

The following example defines a power ground strategy s5 for a standard cell rail connection pattern std_pattern. The nets are Vdd and Vss. The routing area is the core area. The routing layers are M1 and M2 as parameters passed into the pattern. The rails at layer M2 are extended out of the core area by 10 um.

```
prompt> set_pg_strategy \
  s5 \
  -core \
  -pattern {{name: std_pattern}{nets: {Vdd Vss}}{parameters: {M1 M2}}}\
  -extension {{stop: 10}{layers: M2}}
```

MORE EXAMPLES

For more example, please start GUI and invoke the following command.
gui_show_task -task "Design Planning:PG Planning->Examples->Overview"

SEE ALSO

compile_pg(2)
create_pg_composite_pattern(2)
create_pg_macro_conn_pattern(2)
create_pg_mesh_pattern(2)
create_pg_region(2)
create_pg_ring_pattern(2)
create_pg_special_pattern(2)
create_pg_std_cell_conn_pattern(2)
create_pg_wire_pattern(2)
remove_pg_strategies(2)
report_pg_strategies(2)
set_pg_strategy_via_rule(2)
set_pg_via_master_rule(2)

Version V-2023.12-SP5-6
Copyright (c) 2025 Synopsys, Inc. All rights reserved.

Man(1) output converted with man2html